

Artificial Neural Networks (ANNs) are computational systems designed to simulate how the human brain processes information. They are an important concept in **Artificial Intelligence** and are widely used in **Machine Learning** to solve complex problems that traditional programming methods struggle to handle.

An artificial neural network is made up of layers of connected nodes called neurons. These neurons are arranged into three main types of layers: the input layer, hidden layers, and the output layer. The input layer receives data from the environment, such as numbers, text, or images. Hidden layers perform calculations that transform the input into useful representations. The output layer then produces the final result, such as identifying an object in an image or predicting future values.

The strength of the connections between neurons is represented by weights. During the training process, these weights are adjusted so that the network can learn patterns from data. One common training method is **Backpropagation**, which updates weights by minimizing the error between predicted and actual outputs.

Artificial neural networks are applied in many real-world areas, including speech recognition, financial forecasting, autonomous vehicles, and healthcare diagnostics. Modern developments such as **Deep Learning** use very large neural networks with many layers to achieve remarkable performance in tasks like language translation and computer vision.